

LSC Kingwood Fire Science — Water Line & Septic System

CSP 26-08-10 | Materials/quantity breakdown for human estimator review | Report-only, no bid submitted

Report-only. This is a materials/quantity breakdown generated from the real bid attachments for internal review. It does not submit a bid, price a proposal, or commit ENGCO to anything — every line is either cited to an exact source or explicitly flagged as needing human takeoff.

22
total line items

17
cited/verified

5
flagged for human takeoff

DIV	ITEM	QTY	UNIT	CONFIDENCE	SOURCE
33	12-inch AWWA C900 DR18 PVC public water main, blue SOW says 'approximately' with no total LF stated, and the plan set (sheets 04-06) has no bid-quantity summary sheet. Real station callouts exist on the plan-and-profile sheets, but the exact station where 12-inch pipe ends is not printed as text anywhere found — only station tick marks (1+00 to ~5+00) are shown. A precise LF here needs a human with the CAD file or a field-verified station, not a pixel-measured guess from the PDF image (see extract_door_schedule.py's VISION RULE: vision counts/identifies, it does not originate a precise number that belongs in a schedule).	—	lf	unresolved	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — <i>Furnish and install approximately 12-inch AWWA C900 DR18 blue PVC public water main along Kingwood Drive and Sorters Road...</i>
33	Sorters Road segment, 12-inch/8-inch C900 PVC (STA 1+00.00 to STA 4+13.47, tie-in to trenchless crossing) 313.47 LF = 413.47 - 100.00, station arithmetic between two independently, explicitly printed station callouts (both sheets agree on the tie-in station exactly). This is the one segment length on this project defensible enough to call cross_checked rather than a plan-scaled estimate. Includes the 12-inch-to-8-inch transition; sheet 05's profile shows the transition graphically but does not print its exact station.	313.47	lf	cross_checked	26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p5 — STA. 1+00.00 SORTERS ROAD (baseline origin, shared with Kingwood Drive STA 1+00.00) 26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p6 — STA, 4+13.47 SORTERS ROAD = STA. 1+34.68 SORTERS ROAD CROSSING (explicit tie-in station, printed on both sheet 05 and sheet 06)
33	Trenchless 8-inch C900 PVC crossing under Sorters Road Real independent agreement: the SOW's 'approximately 44 linear feet' and the plan's printed '44 LF' label match exactly.	44	lf	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — <i>Construct approximately 44 linear feet of trenchless 8-inch C900 PVC public water line crossing Sorters Road...</i> 26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf

p6 — PROP. 44 LF
TRENCHLESS
CONSTRUCTION
(PUBLIC) —
printed directly on
the Sorters Road
Crossing plan-and-
profile sheet

33	Ductile iron pipe transition sections (AWWA C151, C104 lining, 8-mil poly wrap) Real requirement, shown graphically (hatched) on sheets 04-06 profiles at each vertical-bend/casing location, but no length is labeled at any transition. Needs a human take-off from the profile details.	—	lf	unresolved	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — Provide required transitions between PVC and ductile iron pipe and fittings...
33	12-inch wet connection to existing City water line	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — Perform one 12- inch wet connection to the existing 12-inch City water line on Kingwood Drive... 26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p4 — 1 - 12" WET CONNECTION / REMOVE EXIST. PLUG AND BOV AND CONNECT PROP. 12" WATER LINE TO EXIST. 12" WATER LINE — printed callout, east end of Kingwood Drive sheet
33	Fire hydrant assemblies (6-inch lead, auxiliary valve, thrust restraint) Counted from the plan-view fire-hydrant symbol + station callout on each sheet, not from a schedule table (none exists on this set) — flagged vision_count, not schedule_verified. A human should re-confirm against the full-size plan before pricing.	4	ea	vision_count	26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p4 — Counted 4 fire-hydrant symbols with F.H. callouts across sheets 04-06: STA 1+44 (Kingwood Dr), STA 4+59 (Kingwood Dr), Sorters Rd mid- segment, STA 4+18 (Sorters Rd)

33	Gate valves with box, various sizes (6", 8", 12") at tees/connections Real, plainly labeled on every plan sheet (e.g. '2-12" G.V. W/ BOX', '1-6" G.V. W/ BOX', '1-8" G.V. W/ BOX') at each tee and connection cluster — but several fitting clusters appear on more than one sheet where two sheets share a boundary point (Kingwood Dr/Sorters Rd junction shown on both sheet 04 and sheet 05; Sorters Rd/crossing junction shown on both sheet 05 and sheet 06). Double-counting a shared-boundary cluster is a real risk here — the same failure mode this toolkit already hit and fixed once on Nueces's pier caps. Rather than guess which clusters are duplicates without a clean way to cross-check station coordinates between sheets, this is left for a human to tally directly off the full-size plans (or the source CAD/DWG file, referenced in the plan sheets' title blocks) rather than risk a wrong number with a confident label.	—	ea	unresolved	n/a (unresolved)
33	12-inch fittings: tees, 45° bends, plugs and clamps at line ends/junctions Same shared-boundary double-count risk as the gate valves above — flagged for the same reason, not guessed at.	—	ea	unresolved	n/a (unresolved)
33	8-inch fire-rated compound meter assembly (FRCWMA) with vault, in 31'x15' water meter easement	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — ...8-inch fire-rated compound meter with vault (FRCWMA) in the 31'x15' water meter easement... 26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p6 — PROP. FRCWMA W/ VAULT, labeled on the Sorters Road Crossing plan and profile
33	8-inch reduced pressure zone (RPZ) backflow preventer assembly, on private property	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — ...and the 8-inch reduced pressure zone (RPZ) backflow preventer assembly located on private property... 26 0813 - LSC Kingwood FTF_Approved PUB Plans_260813.pdf p6 — PROPOSED 8" REDUCED PRESSURE ZONE (RPZ) BACKFLOW PREVENTOR IN PRIVATE PROPERTY — key note W2, Sorters Road Crossing sheet
33	System design capacity	272	gpd	schedule_verified	26 0818 SOW Septic System.pdf p1 — The system shall be constructed to accommodate a maximum daily

usage of 272
Gallons Per Day
(GPD).

33	Minimum required trench bottom area	1,360	sf	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>The contractor shall prepare a minimum required trench bottom of 1,360 square feet...</i>
33	Engineered drip disposal area	2,560	sf	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...and install the engineered drip area across 2,560 square feet.</i>
33	Pretreatment tank, 500-gallon	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>Furnish, excavate, and install a 500-gallon pretreatment tank...</i>
33	Clearstream aerobic treatment unit, 800-gallon	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...an 800-gallon Clearstream Treatment Unit...</i>
33	Pump tank, 1,080-gallon SOW explicitly states no chlorinator is required — do not price one.	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...and a 1,080-gallon pump tank. A chlorinator is not required for this specific design.</i>
33	Simplex effluent pump, Goulds 20 EB, 0.5 HP	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...a Simplex Effluent Pump (Goulds 20 EB 1/2 HP)...</i>
33	Solids pump, 2-inch, Goulds, in dosing tank	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...a 2-inch Goulds Solids Pump inside the dosing tank...</i>
33	Effluent pump, Goulds, 0.5 HP, 20 GPM	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — <i>...and a 0.5 HP Goulds 20 GPM Effluent Pump.</i>
33	Dual filter, Aztec, 100 micron	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p2 — <i>Install an Aztec Dual 100 Micron Filter to ensure effluent clarity and protect the drip infrastructure.</i>
33	Drip tubing, Netafim, 0.91 GPH, across the 2,560 SF drip field Quantity re-stated from the drip-area line above — this is the area the tubing must cover, not a separate linear-footage figure (none	2,560	sf	schedule_verified	26 0818 SOW Septic System.pdf p2 — <i>Install Netafim 0.91 GPH</i>

given; drip emitter spacing/lateral spacing needed to convert to LF of tubing is not in the SOW).

drip tubing throughout the designated absorption area.

33 Supply piping, training facility building to septic system (Sch 40 or DWV)

— If

unresolved

No length given in the SOW; the septic system's own 2-page SOW has no site plan showing the building-to-tank distance. Needs the actual site/civil plan for the fire training facility building, which is not included in this bid-attachments package.

26 0818 SOW Septic System.pdf
p1 — *Install Schedule 40 or DWV thick-wall pipe for all connections running from the training facility building to the septic system.*